PSYCHOMETRIKA—VOL. 12, NO. 2
JUNE, 1947

NOTE ON THE SAMPLING ERROR OF THE DIFFERENCE BETWEEN CORRELATED PROPORTIONS OR PERCENTAGES

QUINN MCNEMAR
STANFORD UNIVERSITY

Two formulas are presented for judging the significance of the difference between correlated proportions. The chi square equivalent of one of the developed formulas is pointed out.

It is well known that the sampling variance of the difference between two proportions, p_1 and p_2, based on independent samples is given by

$$\sigma^2_d = \sigma^2_{p_1} + \sigma^2_{p_2} = \frac{p_1 q_1}{N_1} + \frac{p_2 q_2}{N_2}, \tag{1}$$

or by

$$\sigma^2_d = \frac{pq}{N_1} + \frac{pq}{N_2}, \tag{2}$$

in which p_1 and p_2 as separate estimates of the unknown population parameters are replaced by

$$p = \frac{p_1 N_1 + p_2 N_2}{N_1 + N_2}$$

as a better estimate which is consistent with the to be tested null hypothesis, and therefore to be preferred. It is also known that when σ^2_d is calculated by (2) the value of d^2/σ^2_d equals χ^2 with one degree of freedom.

There are many situations in which the sampling variance of the difference between two proportions (or percentages) must take into account the fact that the two proportions are not based on independent samples. For example, p_1 may represent the proportion of individuals who give a particular response prior to a provided experience and p_2 the proportion of the *same* individuals giving the response after the interpolated experience; or p_1 might be the proportion passing one test item and p_2 the proportion of the same group passing a second item, and we wonder whether items differ significantly as to difficulty; or we may wish to test the significance of the difference between the responses of a group to two opinion questions; or we may need to evaluate the difference between two proportions based on paired or matched individuals.

For such situations as the foregoing, neither formula (1) nor (2) is applicable. To be applicable, a subtractive correlational term should be included. The needed correlation is that between the sampling variations of the two proportions. Since it can readily be shown that proportions are means, and since it is known that the correlation between sample means equals that between variates, it follows that the needed correlation is that between first and second responses, or between performances on the two items being compared for difficulty, or between responses to two opinion questions, or between the responses of matched cases. The correlation "scatter plot" for these situations involves a two-by-two table, for which one may apparently choose either the contingency coefficient or tetrachoric r or the fourfold point r as the measure of correlation. Which type of coefficient do we need for the sampling variance of the difference between correlated proportions?

If the answer to this question is to be found in the statistical literature, the writer has failed to locate it. If the answer were obvious, one would not expect the near universal failure of researchers to include the needed correlational term in those situations where it must certainly be taken into account in testing significance. We have found only one text* which considers the correlational term, and the authors of this text, after saying that "it is seldom that we know the correlation factor when dealing with proportions," go on to state without proof that "it would need to be computed by the tetrachoric method." We shall presently see that this idea is incorrect.

The desired sampling variance formula would be of the form

$$\sigma^2_d = \sigma^2_{p_1} + \sigma^2_{p_2} - 2r_{p_1p_2}\sigma_{p_1}\sigma_{p_2}, \tag{3}$$

in which r is not as yet specified. Suppose a fourfold table of frequencies, and the corresponding table of proportions, for a first (I) and a second (II) set of responses from the same individuals (see Table 1). The difference between the two marginal values, $p_2 - p_1$, is the difference for which we seek a standard error. For the sake of exposition, let us assume that this difference represents a change in responses. Then the B and C individuals are those who have not changed, while A represents those who moved from yes to no, and D those who changed from no to yes. The lower-case letters represent the corresponding proportions. Let us score those who changed to

* Peters, C. C. and Van Voorhis, W. R. Statistical procedures and their mathematical bases. New York: McGraw-Hill, 1940. P. 183.

TABLE 1
Fourfold Table of Original and Proportional Frequencies

		(II) No	(II) Yes	
(I)	Yes	A	B	$A+B$
	No	C	D	$C+D$
		$A+C$	$B+D$	N

		(II) No	(II) Yes	
(I)	Yes	a	b	p_1
	No	c	d	q_1
		q_2	p_2	1.00

yes as $+1$, those who changed to no as -1, and those who changed not at all as O. This leads to the frequency distributions for changes shown in Table 2, which includes easily derived expressions for the mean and variance of the distributions of changes.

TABLE 2
Distributions for Changes in Terms of Original and Proportional Frequencies

Original X	Original f	Proportional X	Proportional f
$+1$	D	$+1$	d
0	$B+C$	0	$b+c$
-1	A	-1	a
	N		1.00

$$\text{Mean} = (D-A)/N = d-a = p_2 - p_1$$

$$S.D.^2 = \frac{1}{N^2}[N(D+A) - (D-A)^2] = (d+a) - (d-a)^2$$

Obviously, the mean change equals $(D - A)/N = (d - a) = p_2 - p_1$, or the mean of the changes (or differences) equals the difference between the two proportions. The sampling error of $p_2 - p_1$ must be the same as that for M, its equivalent, and the sampling variance of M is easily obtained by dividing the distribution variance by N. Thus, in terms of proportional frequencies, we have

$$\sigma^2_M = \sigma^2_{p_2-p_1} = \frac{1}{N}[(d+a) - (d-a)^2]. \tag{4}$$

Now by easy, but somewhat cumbersome algebra which need not be reproduced here, it can be shown that (4) reduces to (3) *providing* the r of (3) is the fourfold point correlation coefficient. It will be noted that (4) requires less computation than (3) even when r has been calculated for its own sake.

It will be seen that (3), hence by implication that (4), involves the use of two separate estimates of the unknown population proportions, and therefore both are analogous to (1). If we wish a formula analogous to the preferred (2), that is, one involving an estimate based on the combined sets of responses or a p based on the total as a better estimate, we note that such a p would, as in the case of (2), imply that the population values for the two proportions are equal. This would be consistent with the null hypothesis that $p_2 - p_1 = 0$ except for random sampling fluctuations. The proper p to replace p_2 and p_1 in (3) becomes the simple average of p_2 and p_1 since for the given situation $N_2 = N_1 = N$. Now the null hypothesis that $p_2 = p_1$ also implies that the equivalent, $d - a$, must equal zero except for sampling errors; hence formula (4) becomes

$$\sigma^2_{p_2-p_1} = \frac{1}{N}(d + a) , \tag{5}$$

which is algebraically identical to (3) with p_2 and p_1 replaced by p. Since (2) is associated exactly with χ^2, it is of interest to point out the connection of (5) with χ^2. Table 3 contains the setup for χ^2 for the original observed (O) frequency distribution of Table 2. Since the net

TABLE 3
Schema for χ^2 Test of Changes

X	O	E	$(O-E)^2/E$
$+1$	D	$(D+A)/2$	$\frac{(D-A)^2}{4} \cdot \frac{2}{D+A} = \frac{1}{2}\frac{(D-A)^2}{D+A}$
0	$B+C$		
-1	A	$(D+A)/2$	$\frac{(A-D)^2}{4} \cdot \frac{2}{D+A} = \frac{1}{2}\frac{(D-A)^2}{D+A}$

change (or difference) between the two sets of responses depends upon the D and A individuals, the expected frequencies (E) are written on the assumption that the $D + A$ cases would be divided evenly, on the basis of the null hypothesis, between plus and minus changes. The value of χ^2 becomes the sum of the right-hand terms, or

$$\chi^2 = \frac{(D-A)^2}{D+A} . \tag{6}$$

When we write the square of the critical ratio with (5) as the sampling variance, we have

$$(CR)^2 = \frac{(d-a)^2}{\frac{d+a}{N}} .$$

Replacing proportions by frequencies, this becomes

$$\frac{\left(\frac{D}{N}-\frac{A}{N}\right)^2}{\frac{\frac{D}{N}+\frac{A}{N}}{N}}=\frac{(D-A)^2}{D+A}\ .$$

Thus $(CR)^2$, using formula (5), is identical to the χ^2 of (6). Obviously, $CR=(D-A)/\sqrt{D+A}$, and therefore the significance of the difference between p_2 and p_1 can readily be computed from the fourfold table of frequencies or from the fourfold table of proportions.

The use of either (5) or (6) is to be preferred over (3) or (4) just as, and for the same reason that, formula (2) is preferable to (1).

Although the foregoing development was in terms of changes in responses, the resulting formulas are applicable to the other three situations mentioned at the beginning of this note. The limitation on chi square regarding size of expected frequencies suggests that formulas (6) and (5) should not be used unless $A + D$ is 10 or greater.